

Dealer Markets: a Reinforcement Learning Mean Field Game Approach

M. Bernasconi-de-Luca¹, martino.bernasconideluca@polimi.it, **contact author**
E. Vittori^{1,2}, edoardo.vittori@polimi.it
F. Trovò¹, francesco1.trovo@polimi.it
M. Restelli¹, marcello.restelli@polimi.it

¹ Politecnico di Milano, Dipartimento di Elettronica Informazione e Bioingegneria,
Piazza Leonardo da Vinci, 32, Milano, Italy

² Intesa Sanpaolo, Largo Matteoli 3, Milano, Italy

Dealer Markets: a Reinforcement Learning Mean Field Game Approach

ARTICLE HISTORY

Compiled December 23, 2021

ABSTRACT

We study the problem of finding an equilibrium strategy in an Over The Counter (OTC) market populated by multiple strategic dealers quoting the *bid-ask* prices. The need for an equilibrium strategy comes from the assumption that each dealer adapts their behavior by learning the optimal quoting strategy. Hence, we model the market as a game between many agents competing for resources. Based on this framework, we propose an efficient numerical procedure using Reinforcement Learning and Mean Field Games which learns an approximate equilibrium. Through an experimental campaign, we validate the proposed method in a realistic market-making scenario against strategic dealers trained to exploit our weaknesses and evaluate their performance against non-strategic agents.

KEYWORDS

dealer markets; mean field games; game theory

1. Introduction

In the last decades, financial markets have undergone a process of electronification and automation. The most common market mechanism regulating the exchange of assets is the Continuous Double Auction (CDA) process. While most exchanges work with a CDA mechanism, there is a large part of the Over The Counter (OTC) market operating through Multi-Dealer-to-Client (MD2C) platforms, such as Bloomberg FIT, Tradeweb, or MarketAxess, which are replacing the traditional voice negotiations. The instruments that are traded on MD2C platforms are for instance bonds, swaps and forwards, while CDA mechanisms typically apply to securities, *i.e.*, instruments that can be represented as a certain multiple of a unit fungible contract such as equities, futures, and Exchange Traded Funds (ETFs). Finally, there are dark pools, which are private exchanges used by institutional investors to execute large orders. In this work, we focus on the mostly unexplored MD2C mechanisms and refer to this framework as the *dealer markets* framework.

There are two types of users accessing MD2C platforms: the *dealers* and the *clients*. Dealers, *a.k.a.* market makers, are those who have the mandate to offer liquidity to the market, by continuously quoting *bid* and *ask* prices, and have a business model based on maximizing the transaction volume. Their objective is to realize margin by buying at the bid and selling at the ask price, while mitigating the market risk associated with holding a large inventory of the traded instrument. The clients, *e.g.*, hedge funds, asset managers, and prop traders, have a specific view of the asset and trade to profit from a directional view of the market or for risk management purposes.

In this setting, there are two types of quotes that the dealers can show: *firm*, which means that it is the final price that can be executed; and *indicative*, which instead means that it is an approximate price that the trader may change (slightly) when she/he receives the *Request For Quote* (RFQ) and communicates the firm quote. In the MD2C platforms considered in this work, clients see the indicative quotes of the dealers and send an RFQ, *i.e.*, they request the firm quotes from multiple dealers, as shown in Figure 1a. After that, the client selects the dealer who has published the best price and initiates the trade with her/him, as shown in Figure 1b. An important feature of MD2C platforms is that the firm price of the dealers is seen only by the client and not by the other market makers, which implies that a market maker receives as feedback only the acceptance/rejection of her/his own offer. This process requires strategic behavior among market makers, who want to satisfy the clients' requests while maximizing profits and keeping low risk.

At the same time, always winning the auction process is not desirable for dealers as they are also concerned about keeping their inventory as small as possible. Indeed, keeping the inventory as small as possible is of paramount importance since, after the financial crises, several reforms were introduced in the US, *e.g.*, the Dodd-Frank Act, and in Europe, *e.g.*, Basel III reform, which increased the capital reserve requirements for banks. These reforms impacted the market-making process, by increasing the penalty of holding an inventory, which thus became less profitable for market makers. This lower profitability has forced some banks out of the sector and others to cut costs by automating the process of quoting prices. The lower liquidity in certain assets also paved the way for other financial entities, such as prop trading firms, to enter the market making space using quoting algorithms.¹ The goal of this paper is to explore a method for designing autonomous market makers which exploit Reinforcement Learning (RL) techniques to learn effective market-making strategies even when the other market participants are strategic dealers.

A common market-making strategy that can be easily executed by a human operator requires looking at the average market spread, *e.g.*, the CBBT bid/ask spread, and offering a lower ask for positive inventory, and, vice-versa, offering a higher bid for negative inventory. Conversely, more advanced strategies use a hard-coded pricing strategy for RFQs of small sizes and send to human traders the most relevant orders. The arrival time and the characteristics of each RFQ depend on a variety of factors, including the general market sentiment, the specific asset in each asset class being considering, and the time of day. Nonetheless, even with thorough knowledge of the future RFQs, defining a strategy to optimally perform the market maker's task is a rather complex problem. Indeed, the rewards obtained by each market maker depend not only on his/her actions, but also on the actions of all the other market makers involved in the exchange. This mutual dependence among market makers makes it necessary to model this environment in a *game theory* setting, where multiple rational players compete to maximize their gain over time.

However, the naïve formulation of such a framework using a classical game-theoretical fashion results in a computationally intractable N -player stochastic game. In this work, we propose the use of the Mean Field Games (MFGs) framework, defined by Huang et al. (2006) to model the dealer market framework and find an approximate solution to the original problem. Indeed, this approach is able to efficiently approximate an N -player stochastic game and offers theoretical guarantees on convergence to an equilibrium. The idea of using MFGs to solve such a problem was inspired by the

¹<https://corporatefinanceinstitute.com/resources/knowledge/trading-investing/proprietary-trading/>.

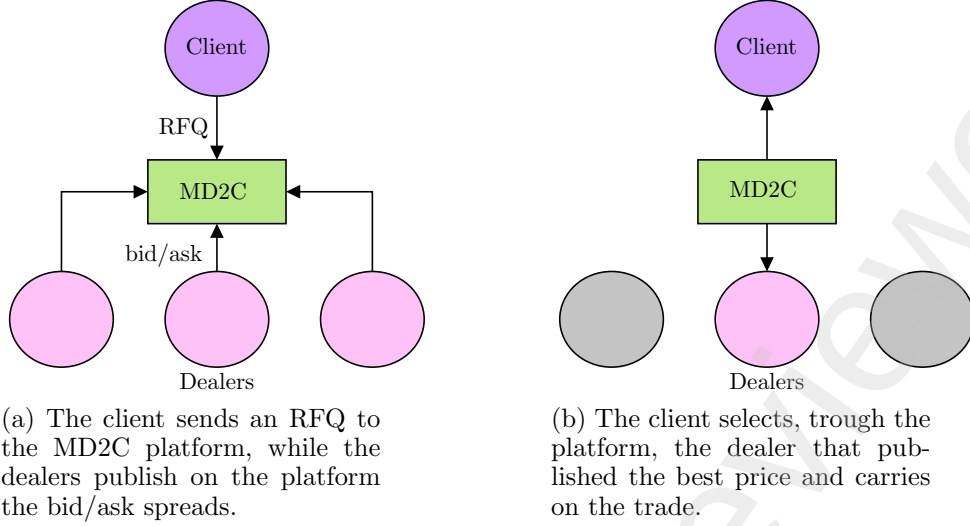


Figure 1.: Schematic representation of the interactions occurring in a MD2C platform.

works of Iyer et al. (2014); Balseiro et al. (2015), which have shown promising results in modeling and solving other auction-type problems. Indeed, the proposed formulation is of crucial importance for practical applications, since the existing approaches in the literature assume the availability of data on the behavior of other dealers, which are commonly private information and not disclosed to competitors.

Novel Contribution In this paper, we provide a method to learn the joint equilibrium strategy of all the players in dealer market. Motivated by the fact that finding an analytical solution to MFGs in general cases is an open problem, we model the dealer market environment as a multi-agent game and solve it using model-free Reinforcement Learning (RL) on discrete-time MFGs (Gomes et al. 2010). Our framework is extremely flexible; it is an approach where the equilibrium is learned by self-play for any model that governs the exogenous stochasticity of the problem, *e.g.*, the price of the asset, and/or the arrival times of the RFQs. Finally, we analyze the behavior of our approach with an extensive experimental campaign where different trading agents compete against each other and draw important conclusions from the experimental campaign, shedding light on the inner working of real dealer markets.

Paper Structure The paper is structured as follows. In Section 2, we describe and comment on existing approaches that model the market maker paradigm, focusing on the ones using an RL approach. Then, we outline the works that define the MFG theory. In Section 3, we specify how the market-making problem can be modeled as a stochastic game, highlighting the most relevant modeling assumptions required. In Section 4, we describe the approach we develop to solve the market-making game and, finally, in Section 5, we provide an experimental evaluation both in the case of a market where there is a strategic adversarial opponent and in the case of a market populated by different dealer configurations, which are commonly used in the literature.

2. Background

The proposed framework unifies two different streams of literature: the former focused on modeling dealer markets, and the latter on model-free MFGs. To the best of our knowledge, our work is the first to use MFGs to model the dealer market scenario. In what follows, we provide a description of the existing literature on these two topics starting with existing market-making approaches, focusing on those that either model the MD2C platforms or use reinforcement learning approaches to solve them. In the following section, we will describe the current literature on MFGs, focusing on those that have taken a model-free approach.

2.1. Dealer Markets

The most common market-making approach is that by Avellaneda and Stoikov (2008), which revived old papers by Ho and Stoll (1981, 1983). After this seminal paper, several other works have been proposed (Cartea et al. 2014; Guéant et al. 2013; Guéant 2016, 2017; Guéant and Manziuk 2019). The most important assumption is that market prices are given by stochastic processes exogenous to the market maker’s behavior, commonly geometric Brownian motion, and that the outcome of an auction depends on a stochastic process that is assumed to model the behavior of other market participants. The closest to our approach is by Guéant and Manziuk (2019), where they use a deep RL approach, i.e., a model-based actor-critic-like algorithm, to find the optimal solution in discrete time. Nonetheless, even in this work, the probability of winning the RFQ and thus the other player’s behavior is modeled as a stochastic process. Indeed, all the above-mentioned works are starting from the rather strong assumption that the market makers are behaving stochastically, even though in reality dealers are able to adapt over time to the strategy of other market makers. Conversely, our approach models multiple competing market makers as opponents in a game, thus requiring no assumption about the behavior of other market makers.

The modeling of multiple competing market makers can be naturally cast in a multi-agent formulation, which provides a model closer to reality. Indeed, multi-agent modeling of the markets has already been used to describe the behavior of the CDA process, for example, in (Darley et al. 2000; Das 2005, 2008), to simulate data in (Byrd et al. 2019), and to analyze liquidity in the dealer market framework in (Bank et al. 2021). However, the only work that uses a multi-agent approach to model market makers in an MD2C platform is by Ganesh et al. (2019), which considers M competing market maker agents. Out of all the mentioned papers, the latter is the most closely related to our work as they also apply RL techniques to optimize their agent’s payoff in a multi-agent framework. However, unlike our approach, the other market makers are hard-coded and fixed, so they cannot behave in an adversarial way. The risk is that the proposed solution might suffer from large losses when deployed in a real-world setting, competing against real market makers who differ in behavior from what was hypothesized during the training. Indeed, one would prefer to have an agent that learns a policy that works against *all* possible behaviors of other players.

Finally, there are other approaches that consider RL in a market-making scenario, but they focus on the CDA auction process, these include (Chan and Shelton 2001; Lim and Gorse 2018; Spooner et al. 2018; Spooner and Savani 2020), which is inherently different from the MD2C markets.

2.2. Mean Field Games

The notion of MFGs was inspired by the economic literature, studying topics related to financial markets (Aumann 1964). MFGs are designed to learn an approximate *Nash equilibrium* in stochastic games with a large number of identical agents, where the equilibrium is expressed as a policy/state measure pair. Specifically, under the given state measure, the policy should be optimal and when the agent applies this policy, the resulting distribution of the agent state is the same as the state measure. In such a framework, all players are identical, anonymous, and have symmetrical interest, thus, the learning problem can be reduced to characterizing the optimal interactions between a reference player and the population, which in its turn behaves as the reference player. In this context, with continuous-time interactions, the Nash equilibrium is commonly computed by providing the solution of a coupled system of dynamical equations in which the first equation models the forward dynamics of the population distribution, and the second one consists of the dynamic programming equation of a reference player.

This framework has been introduced by Huang et al. (2006); Lasry and Lions (2007), who showed the existence of an approximate Nash equilibrium in continuous-time MFGs. Subsequently, several different models of continuous-time MFGs have been proposed: Huang et al. (2007); Huang (2010) consider linear-quadratic games, Saldi et al. (2019, 2020); Tembine et al. (2013) consider risk-sensitive MFGs, while Moon and Başar (2016) study games with Markov jumps. Other works focus on finding numerical approximations to efficiently solve the system of dynamical equations using finite difference methods (Achdou and Capuzzo-Dolcetta 2010; Achdou et al. 2012), semi-Lagrangian schemes (Carlini and Silva 2014, 2015), primal-dual methods (Briceno-Arias et al. 2018; Briceño-Arias et al. 2019), neural network approximations (Carmona and Laurière 2019; Fouque and Zhang 2020; Ruthotto et al. 2020), generative adversarial networks (Cao et al. 2020; Lin et al. 2020), and fictitious play schemes (Robinson 1951; Cardaliaguet and Hadikhannloo 2017; Perrin et al. 2020). An overview of MFGs can be found in the survey by Gomes et al. (2014), or in the books by Fudenberg et al. (1998); Bensoussan et al. (2013).

Conversely, discrete-time MFGs, as required by the dealer market scenario, have not been studied as much. Gomes et al. (2010); Adlakha et al. (2015); Saldi et al. (2019) establish the existence of a mean-field equilibrium without proposing algorithms capable of converging to the equilibrium. The problem of finding a solution, i.e., an equilibrium, in the discrete-time MFGs setting, has been successfully tackled by using RL algorithms. For instance, Fu et al. (2019) develop an actor-critic algorithm learning the equilibrium for mean field control, Carmona et al. (2019a) establish the convergence of a policy gradient algorithm, Carmona et al. (2019b); Anahtarçı et al. (2019) establish the convergence of Q-learning for deterministic systems. Finally, Yang et al. (2018a,b) apply classical RL algorithms to compute the mean-field equilibrium. Among the aforementioned works, of particular relevance for this paper is that by Anahtarçı et al. (2019) which proposes the use of Q-learning and Fitted Q iteration (FQI). They give convergence guarantees with Q-learning, approximation error bounds with FQI, and also analyze the approximate case of a finite number of players.

In the classical MFG framework used in the aforementioned works, the rewards and the dynamics for each player are known. Specifically, they depend only on the state/action pair of the player and the population state distribution. Conversely, in the market-making problem, the reward and the dynamics for each player also depend on the actions of all the other players. Therefore, the more inclusive framework of

General Mean Field Games (GMFGs) by Guo et al. (2019, 2020) is required to handle this additional complexity faced by the market-making scenario. Note that the analyses provided in the cited works are merely theoretical and do not provide any constructive solution for finding an equilibrium. Furthermore, they are general formulations of GMFG, while the application of this framework to the problem of dealer markets has not, to the best of our knowledge, been explored before.

3. Model (Stochastic game)

Let $\{\mathcal{F}_t\}_t$ be the natural filtration generated by the market information available up to time t . The market maker is in charge of an asset whose price P_t is an $\mathcal{F}_t$ -adapted process. At each time $t \in \mathbb{N}$, a client can submit an RFQ to the MD2C platform, specifying the volume v_t (that is $\mathcal{F}_t$ -adapted), where its sign specifies the side of the trade (positive or negative to manifest the intention to acquire or sell an asset, respectively). The MD2C platform is available to a large number N of market makers, even though at each time t only a number $M_t < N$ (possibly stochastic and $\mathcal{F}_t$ -adapted), a.k.a. *market thickness* (Iyer et al. 2014), will respond to the RFQ. Each market maker i responds to the RFQ by showing the client the prices $(P_{t,buy}^i, P_{t,sell}^i)$, which characterize the firm price at which the client can buy or sell the asset when market maker i is selected for the trade. We assume that the spreads published by the market maker are a function of the volume v_t communicated by the client. The selected market maker i will buy from the client at price $P_{t,buy}^i(v_t)$ and will sell to the client at price $P_{t,sell}^i(v_t)$. Given the quotes from all the M_t market makers who answered the RFQ at time t , the client will select the market maker i^* giving the best quote for the placed RFQ, i.e., the smallest spread in the direction indicated by the RFQ. We will denote by z_t^i the inventory of dealer i at time t .

N-player Stochastic Games Let us define a discrete-time N -player Markovian stochastic game. This model is an extension of Markov Decision Processes (MDPs), where the transition probabilities depend on the aggregated action of all players and the reward depends on the state of all players. At any time $t \in [N]$, each agent is in state $s_t^i \in \mathcal{S}_i$ and chooses an action $a_t^i \in \mathcal{A}_i$.² Define $\mathcal{S}^N := \mathcal{S}_1 \times \dots \times \mathcal{S}_N$ as the set of possible joint states and $\mathcal{A}^N := \mathcal{A}_1 \times \dots \times \mathcal{A}_N$ as the set of possible joint actions performed by players. Given the current joint state $\mathbf{s}_t := (s_t^1, \dots, s_t^N) \in \mathcal{S}^N$, and current joint action $\mathbf{a}_t = (a_t^1, \dots, a_t^N) \in \mathcal{A}^N$, each player $i \in [N]$ receives a reward specified by the reward function $r^i : \mathcal{S}^N \times \mathcal{A}_i \rightarrow \mathbb{R}$. Furthermore, the current state s_t^i of each player evolves into the next state s_{t+1}^i according to the transition function:

$$s_{t+1}^i \sim \mathcal{P}(\cdot | s_t^i, \mathbf{a}_t) \quad \forall i \in \mathcal{N}. \quad (1)$$

Similarly to the case of MDPs, the policy $\pi^i(\mathbf{s}_t)$ of an agent i is a function that maps a state $\mathbf{s}_t$ to a distribution over the action space action $\Delta(\mathcal{A}_i)$, formally $\pi^i : \mathcal{S}^N \rightarrow \Delta(\mathcal{A}_i)$. We denote by π^{-i} the policies of all players except the i -th one, so that $\pi = (\pi^i, \pi^{-i})$. The expected cumulative reward of agent i following a generic policy

²Given $W \in \mathbb{N}$, with $[W]$ we denote the set $\{1, \dots, W\}$.

sequence π is:

$$J_{\pi}^i := \mathbb{E}_{\substack{\mathbf{a}_t \sim \pi(\mathbf{s}_t), \\ \mathbf{s}_{t+1}^i \sim \mathcal{P}(\cdot | \mathbf{s}_t^i, \mathbf{a}_t)}} \left[\sum_{t=1}^{+\infty} \gamma^t r^i(\mathbf{s}_t, \mathbf{a}_t^i) \middle| s_0 = s \right]. \quad (2)$$

When the policy π^{-i} of the other agents is fixed, the objective of each agent $i \in [N]$ is to solve the following stochastic optimization problem:

$$\pi^i = \arg \max_{\pi} V_{(\pi, \pi^{-i})}, \quad (3)$$

where $V_{\pi}^i := \mathbb{E}_{s \sim \xi} [J_{\pi}^i(s)]$, for an initial state distribution ξ .

Note that the modeling provided by N -player stochastic games is able to describe the dealer markets problem. In what follows we will show the correspondence between the two.

Reward The reward obtained by an agent can be decomposed into three parts: the gain in having bought (resp. sold) lower (resp. higher) than the mid (average of the best bid and the best ask) which we call spread PnL; the profit/loss given by the change in the market price that we call inventory PnL and a penalty for holding the inventory. Formally, the reward for each market maker i is defined as:

$$r_t^i = \underbrace{v_t(P_{t,h}^i(v_t) - P_t)\mathbb{I}_{i=i^*}}_{\text{spread PnL}} + \underbrace{z_{t-1}^i(P_t - P_{t-1})}_{\text{inventory PnL}} - \underbrace{\phi(z_t^i)}_{\text{inventory penalty}}, \quad (4)$$

where $\phi : \mathbb{R} \rightarrow \mathbb{R}^+$ is a function encoding the penalty of owning a net inventory. The definition of reward provided in Equation (4) is common in the literature (Guéant and Manziuk 2019; Avellaneda and Stoikov 2008; Cartea et al. 2015; Guéant 2017), where the risk-aversion of a market maker is encoded implicitly in the function $\phi(\cdot)$. Notice that it is possible to add an explicit penalization for the variation of the rewards over time by making the reward r_t^i dependent on the variability of the past inventory.

Actions The action space $\mathcal{A}_i$ for market maker i is given by the prices $(P_{t,buy}^i(v), P_{t,sell}^i(v))$. Here, we choose to follow a modeling approach similar to the one proposed by Ganesh et al. (2019). Defining the action space $\mathcal{A}_i = [-\epsilon, \epsilon]^2$ for all $i \in \mathcal{N}$, with $\epsilon > 0$, the pricing function is defined as follows:

$$P_{t,h}^i(v) = \tilde{P}_{t,h}(v)(1 + \epsilon_h^i), \quad (5)$$

where $h \in \{buy, sell\}$, $\epsilon_h^i \in [-\epsilon, \epsilon]$ is the action chosen by the market maker i , and $\tilde{P}_{t,h}(v)$ is a reference price calibrated on the market.

State The state space is $\mathcal{S} = [-Z_m, Z_m] \times [P_l, P_u]$, where $Z_m > 0$ is the maximum inventory allowed to be held, and $0 < P_l < P_u$ are the minimum and maximum values for the asset price in consideration, respectively.

Transition Function The transition function $\mathcal{P}(\cdot|s_t, \mathbf{a}_t)$ of the MDP specifies the evolution of the state $s_t = (z_t, P_t)$ to the next state $s_{t+1} = (z_{t+1}, P_{t+1})$ for each player. Specifically, the asset price P_t evolves accordingly to a transition function that is common to all players, while the inventory z_t^i evolves differently for the winner of the auction. Formally, the z_t^i dynamic is described by the following equation:

$$z_t^i = z_{t-1}^i + v_t \mathbb{I}_{i=i^*}, \quad (6)$$

where i^* is the index of the dealer who provided the best quotes.

The objective of each dealer i is to find a policy π^i specifying, for each combination inventory and price in $\mathcal{S}$, the optimal bid/ask spreads in $\mathcal{A}_i$ that maximize their objective function in Equation (3).

Remark 1. The main appeal of the proposed framework is that it allows modeling various features of interest to the dealer. For instance, one could be interested in including the relationship between the distribution of the RFQs and the price process of the underlying asset in the model, i.e., simulating a market sell-off with many bid RFQs but no ask RFQs.

Remark 2. Notice that, the information required to run this scheme is easily available to dealers. Indeed, to fit the parameters of the RFQ arrival process, one needs only the information collected by interacting with the MD2C platform (such as the distribution of the volume v_t , the price P_t and the reference market price $\tilde{P}_{t,h}(v)$). Conversely, other existing approaches make use of information about the behavior of the other dealers, e.g., the work by Fermanian et al. (2016), which is in general hard to obtain as it constitutes the private information of competitors.

4. Learning Equilibrium via Model-Free General mean-field Games

The model described in the previous section provides an accurate description of a real dealer market. However, modeling this environment as an N -player stochastic game suffers from the combinatorial complexity w.r.t. the number N of players (Daskalakis et al. 2009). In what follows, we describe the mean-field Game approach which allows us to obtain an approximate equilibrium strategy of the game.

General mean-field Game approximation of the Dealer Market problem

The assumptions of the mean-field approximation we resort to are that market participants are homogeneous and the number of players N goes to infinity while maintaining a finite market thickness M_t . In practice, under this approximation, the players' population is compactly represented by a distribution on the state-action space $\mathcal{L} \in \Delta(\mathcal{S} \times \mathcal{A})$ and the interaction between each individual player and the rest of the distribution occurs only through the mean-field $\mathcal{L}$. At each time t and for each RFQ, the market participants answering the RFQ are generated by sampling $M_t - 1$ players (market makers) from the mean-field $\mathcal{L}$, thus specifying their state s^k and action a^k for $k \in [M_t - 1]$. In the next paragraph, we will describe formally the GMFG framework, which expands and generalizes MFGs.

Notice that assuming a homogeneous policy between the dealers is only an assumption needed to find an approximate equilibrium, while it does not represent a simplifying assumption with respect to a real market-making framework. Indeed, we will

show in the experimental campaign that the agents trained with the MFG approach provided here maintain good performance (in terms of profit/volatility tradeoff) even when tested against agents with different policies, i.e., in settings where the homogeneous assumption does not hold.

General mean-field Games General mean-field Games (GMFG) are an extension of the mean-field Games introduced by Guo et al. (2019), which allows the next state transition and reward to also depend on all the actions played by the population. Let the *mean-field* $\mathcal{L}_t$ be a distribution of the population over the state-action space, or, formally, $\mathcal{L}_t \in \Delta(\mathcal{S} \times \mathcal{A})$. At each time t , a player in state $s_t \in \mathcal{S}$ plays the action $a_t \in \mathcal{A}$ and receives the reward distributed according to $r(s_t, a_t, \mathcal{L}_t)$. Consequently, the player's current state evolves to s_{t+1} , according to the distribution $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t, \mathcal{L}_t)$. In such a modeling approach, the definition of the value function, given a mean-field $\mathcal{L}$ is:

$$V(\pi, \mathcal{L}) := \mathbb{E}_{\substack{a_t \sim \pi_t(\cdot | s_t, \mu_t), \\ s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t, \mathcal{L}_t)}} \left[\sum_{t=1}^{+\infty} \gamma^t r(s_t, a_t, \mathcal{L}_t) \right], \quad (7)$$

where we defined $\mu_t \in \Delta(\mathcal{S})$, i.e., for each $s \in \mathcal{S}$ we define $\mu_t(s) := \int_{\mathcal{A}} \mathcal{L}(s, a) da$ as the marginal distribution of the population $\mathcal{L}$ over the state space $\mathcal{S}$.

The definition of a Nash Equilibrium profile for a GMFG is defined as follows:

Definition 4.1. The tuple $(\pi^*, \mathcal{L}^*)$ is a Nash Equilibrium for the GMFG if, for any policy π , it holds that:

$$V(\pi^*, \mathcal{L}^*) \geq V(\pi, \mathcal{L}^*), \quad (8)$$

and

$$a_t \sim \pi_t^*(\cdot | s_t, \mu^*), \quad s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t, \mathcal{L}^*), \quad (9)$$

where μ^* is the marginal distribution of the mean state-action population $\mathcal{L}^*$.

To find the equilibrium strategy π^* , we employ the algorithm proposed by Guo et al. (2019). The high-level structure of the algorithm is presented in Algorithm 1. The main idea is to alternately evolve the policy π_w and the mean-field $\mathcal{L}_w$ for $w \in [W]$. Specifically, the algorithm requires an initial mean-field $\mathcal{L}_0$, an environment simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, and the number of iterations W . The simulator $\mathcal{E}(\cdot, \mathcal{L}_w)$ is a function that simulates the environment step, or, formally, for a mean-field $\mathcal{L}$, we have that $(s_{t+1}, r_{t+1}) = \mathcal{E}(s_t, a_t; \mathcal{L})$. For W steps, the algorithm learns an (approximate) optimal policy π_w , by fixing the mean-field distribution $\mathcal{L}_w$ (Line 2).³

After that, during the update of the mean-field $\mathcal{L}_w$, the optimal policy π_w is used to generate a new mean-field distribution $\mathcal{L}_{w+1}$ (Line 3). At the end of the process, the algorithm returns the tuple $(\pi_W, \mathcal{L}_W)$, which is guaranteed to converge to an approximate Nash equilibrium for the GMFG (Guo et al. 2020).

³The number of iterations W should be selected so that the error on the Weierstrass distance between $\mathcal{L}_{W-1}$ and $\mathcal{L}_W$ is smaller than a required threshold. See the work by Guo et al. (2020) for details.

Algorithm 1 Model Free GMFG

Require: Initial mean-field $\mathcal{L}_0$, environment simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, number of iterations W

- 1: **for** $w \in [W]$ **do**
- 2: Find the single-agent optimal policy π_w with fixed mean-field $\mathcal{L}_w$
- 3: Update $\mathcal{L}_{w+1}$ using $\mathcal{E}(\cdot, \cdot; \mathcal{L}_w)$
- 4: **end for**
- 5: Return $(\pi_W, \mathcal{L}_W)$

GMFG with FQI In what follows, we present the FQI for GMFG algorithm, whose pseudo-code is provided in Algorithm 2, an implementation of Algorithm 1 in which the FQI algorithm (Ernst et al. 2005; Antos et al. 2007) is used to find the policy π_w . Specifically, the procedure described in Line 2 of Algorithm 1 corresponds to Lines 2:7 of Algorithm 2, while the update of the mean-field $\mathcal{L}_w$ of Line 3 of Algorithm 1 corresponds to Lines 8:14 of Algorithm 2.

More precisely, the algorithm requires as input an initial mean-field $\mathcal{L}_0$, a simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, and the parameters W , J , I , and τ . Specifically, W is the number of times the FQI algorithm and the evolution of the mean-field $\mathcal{L}_w$ are performed, J represents the number of iterations to be set in the FQI algorithm, I is the number of samples to generate the empirical distribution for $\mathcal{L}_w$ and τ is the inverse temperature parameter to be used in a Boltzmann distribution.

For each of the W iterations, the FQI algorithm is employed to find the policy π_w . Indeed, the goal of the FQI algorithm is to approximate the state-action function $Q(s, a)$, i.e., the cumulative value of a specific action a in a state s , by means of a regressor $\hat{Q}_w(s, a)$ that exploits the dataset $\mathcal{D}_w$. At first, for each state $s \in \mathcal{S}$ and action $a \in \mathcal{A}$, the algorithm initializes the approximated state-action function $\hat{Q}_{w,0}(s, a)$ to zero (Line 2) and generates, using the current environment simulator $\mathcal{E}_{\mathcal{L}}$, a dataset:

$$\mathcal{D}_w := \{(s_i, a_i, r_i, s'_i)\}_{i \in [D]}, \quad (10)$$

where, for the i -th sample, s_i is the current state, a_i is the action performed, $r_i = r(s_i, a_i, \mathcal{L}_w)$ is the realization of reward received from the environment with a mean-field $\mathcal{L}_w$, and s'_i is the next state (line 3). Subsequently, the algorithm solves a regression problem having as target $r_i + \gamma \min_{a \in \mathcal{A}} \hat{Q}_{w,j}(s', a)$ (Line 5), i.e., the currently available estimate of the value of the action a in state s .⁴ The regression step is repeated J times (Line 4), as prescribed by the FQI algorithm, and results in an approximation $\hat{Q}_{w,J}(s, a)$ of the true state-action function. From the state-action function, we extract the approximate optimal policy $\pi_w(\cdot)$ for each state $s \in \mathcal{S}$ by applying the soft-max function to $\hat{Q}_{w,J}(s, a)$ (Line 7). This ensures that $\pi_w(s)$ defines a probability distribution over the actions $\mathcal{A}$ for a specific state s .

In the pseudo-code following Line 7, the algorithm performs an update of the population $\mathcal{L}_w$ due to the change in the players' policy $\pi_w(\cdot)$. First, it computes the marginal distribution of the population over the state space μ_w (Line 8). After that, it samples

⁴In what follows we used as hypothesis space for the regression problem the class $\mathcal{F}$ of randomized decision tree regressor (Ho 1995), due to their robustness in performances with respect to the choice of the hyperparameters. Depending on the computational complexity required by this step and the characteristics of the environment, one might select a different regressor, e.g., Support Vector Machines or Neural Networks. See (Antos et al. 2007) for details.

Algorithm 2 FQI for GMFG

Require: Initial state-action distribution $\mathcal{L}_0$, environment simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, number of iterations I , W , and J , inverse temperature parameter τ

```
1: for  $w \in [W]$  do
2:   Initialize  $\hat{Q}_{w,0}(s, a) \leftarrow 0 \forall s \in \mathcal{S}, a \in \mathcal{A}$ 
3:   Generate dataset  $\mathcal{D}_w = \{(s_i, a_i, r_i, s'_i)\}_{i \in [D]}$  using simulator  $\mathcal{E}(\cdot, \cdot; \mathcal{L}_w)$ 
4:   for  $j \in [J]$  do
5:      $\hat{Q}_{w,j+1} = \arg \min_{f \in \mathcal{F}} \sum_{i \in D} \left( f(s_i, a_i) - r_i - \gamma \max_{a \in \mathcal{A}} \hat{Q}_{w,j}(s'_i, a) \right)^2$ 
6:   end for
7:   Extract  $\pi_w(s) = \frac{\exp(\tau \hat{Q}_{w,J}(s, \cdot))}{\sum_{a \in \mathcal{A}} \exp(\tau \hat{Q}_{w,J}(s, a))} \forall s \in \mathcal{S}$ 
8:    $\mu_w \leftarrow \int_{\mathcal{A}} \mathcal{L}_w(s, a) da$ 
9:   Initialize  $\mathcal{L}_{w+1}(s, a) \leftarrow 0 \forall s \in \mathcal{S}, a \in \mathcal{A}$ 
10:  for  $i \in [I]$  do
11:     $s_i \sim \mu_w, a_i \sim \pi_w(s_i)$ 
12:     $(s'_i, r'_i) \leftarrow \mathcal{E}(s_i, a_i; \mathcal{L}_w)$ 
13:     $\mathcal{L}_{w+1}(s'_i, a_i) \leftarrow \mathcal{L}_w(s'_i, a_i) + 1/I$ 
14:  end for
15: end for
16: Return  $\pi_W$  and  $\mathcal{L}_W$ 
```

I states and actions pairs from the distributions μ_w and π_w , respectively, (Line 11) which are used in the simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L}_w)$ to generate the next state s'_i (Line 12). Finally, it updates the empirical distribution of the population $\mathcal{L}_w(s'_i, a_i)$.

The above two steps are repeated W times, after which the algorithm returns an estimate of the approximate optimal policy π_W and the mean-field $\mathcal{L}_W$.

Remark 3. It is important to note that the infinite player approximation still provides theoretical guarantees to the original stochastic game, where the number of players N to whom the platform is available is finite. In particular, if we have an equilibrium for the GMFG, then it is an ε -Nash equilibrium for the finite player game with $N = N(\varepsilon)$ players. An ε -Nash equilibrium for an N -player stochastic game, is a collection of N policies $(\pi_1^*, \pi_2^*, \dots, \pi_N^*)$ such that for all players $i \in [N]$:

$$V(\pi_i^*, \pi_{-i}^*) \geq V(\pi, \pi_{-i}^*) - \varepsilon, \forall \pi, \quad (11)$$

with $\varepsilon > 0$. For reference, see Theorem 2.1 in (Anahtarci et al. 2019).

Remark 4. Note that Guo et al. (2019) proposed a solution that, instead of using FQI for the optimal policy estimation, uses the Q -learning algorithm (Watkins and Dayan 1992).⁵ However, this approach does not perform well on the problem of dealer markets, as it models the state-action function $Q(\cdot, \cdot)$ in a tabular fashion, thus not exploiting its regularity structure. Conversely, using the Q -learning version which uses a functional approximation for the Q -function (e.g., DQN (Mnih et al. 2015)) would loose the theoretical guarantees on convergence to the equilibrium.

⁵See Algorithm 3 in Appendix A for details.

5. Experimental Campaign

In what follows, we provide the experimental results achieved with the proposed framework. At first, we introduce the model of the environment, i.e., the evolution of the price, the reference spread, and the RFQ process used in the experiments. Then, we present the different dealer agents who will interact in the market-making simulations. Finally, we present the relevant metrics employed in evaluating our method.

After explaining the setting, we provide empirical simulations for two different dealer market environments. In the former, we explore the robustness of the equilibrium policy by testing it in a market where one player has knowledge over the strategy of the other player. In the latter, we provide a wide experimental campaign in which we populate the market with many players with different strategies, strategic and/or stochastic, and verify how they interact over time.

5.1. Experimental Setting

Synthetic environment The price process P_t is modeled as a geometric Brownian motion, formally:

$$P_{t+dt} := P_t \exp \left\{ \left(\nu - \frac{\sigma^2}{2} \right) dt + \sigma Z_t \sqrt{dt} \right\}, \quad (12)$$

where $Z_t \sim \mathcal{N}(0, 1)$ is a normal random variable, i.e., zero mean, unit variance, dt is the time interval, $\nu \in \mathbb{R}$ is the mean return over the time dt , and $\sigma > 0$ is the volatility of the asset. These values have been annualized as $dt = 1/250$ corresponding to one RFQ per day over a year, $\nu = 0$ to ensure the price process is a martingale, and $\sigma = 0.2$ as a common value for the volatility of an equity stock (Barles and Sonner 1998). The volume v_t of the RFQ at time t is extracted from $\{-1, 1\}$ with equal probability. The possible buy and sell actions for each market maker are in $\mathcal{A} = \{-0.03, -0.02, -0.01, 0, 0.01, 0.02, 0.03\}$ which correspond to the typical tick-sizes of the equity markets. The inventory penalization of the reward defined in Equation (4) has been set to $\phi(z) = \frac{z^2}{2}$, as commonly done in the literature (Ganesh et al. 2019; Guéant and Manziuk 2019). Finally, following what has been done in (Kolm and Ritter 2019), we defined the market spread as:

$$\tilde{P}_{t,b}(v) = P_t + \delta(|v| + 0.01v^2), \quad (13)$$

where $\delta = 0.1$.

MFG Agents We trained two FQI-GMFG agents: FQI_2 and FQI_4 . FQI_2 was trained considering a market thickness of $M_t = 2$, for each $t \in [T]$, while FQI_4 considered a market thickness of $M_t = 4$, for each $t \in [T]$. In both cases, we fixed $W = 5$, $J = 5$, $I = 2 \cdot 10^6$, $\tau = 4$, and $\gamma = 1$. As a regressor we used extra-trees (Ernst et al. 2005) with 500 trees and a min-split parameter of 0.001, whose parameters have been selected on preliminary experiments.

Benchmark agents We compared our solution with agents following stochastic strategies and with solutions from the literature (e.g., in Ganesh et al. (2019)). Specifically, we analyzed:

- Persistent ($\mathfrak{P}$) agents, which select the action $(\epsilon_b, \epsilon_s) = (0, 0)$ for each state $s \in \mathcal{S}$;
- Uniform ($\mathfrak{U}$) agents, which randomly select an action $(\epsilon_b, \epsilon_s) \in [-\epsilon, \epsilon]^2$ for each state $s \in \mathcal{S}$;
- Normal ($\mathfrak{N}$) agents, which randomly select an action $\epsilon_b \sim \mathcal{N}(0, 0.04)$, and $\epsilon_s \sim \mathcal{N}(0, 0.04)$, i.e., from Gaussian distributions.⁶
- Q-learning agents (Q_2, Q_4), generated using the Q-GMFG algorithm described in Appendix A. We trained two agents Q_2 , and Q_4 , for market thickness $M_t = 2$, for each $t \in [T]$, and $M_t = 4$, for each $t \in [T]$, respectively.

Metrics We evaluate the strategies in terms of:

- mean dollar reward: $L := \sum_{t \leq T} \frac{l_t}{T}$, where the loss l_t for each agent i is defined as $l_t := \mathbb{I}_{i=i^*} |v_t(P_t, (v_t) - P_t)| + z_{t-1}(P_t - P_{t-1})$;
- mean Sharpe ratio on the dollar reward: $S := \frac{L}{\sqrt{\sum_{t \leq T} \frac{(l_t - L)^2}{T}}}$;

where the length of each run is $T = 500$ days.

Note that we study two different metrics to evaluate different aspects of each strategy. Specifically, we want to evaluate the average dollar return provided by a strategy and its reliability.

5.2. Equilibrium Policy

As a preliminary analysis, we investigate the behavior of the policy π_W learned by the FQI-GMFG agents in the environment described in the previous section. In particular, we investigate the policy π_W as a function of the inventory z_t . For each inventory $z \in [-Z_m, Z_m]$, we generated 100 states by sampling uniformly a price $P_k \in [P_l, P_u]$, registered $(\epsilon_{buy}, \epsilon_{sell}) = \pi_W(z, P_k)$ for each state (z, P_k) , and averaged over the 100 samples. Figure 2 shows the average strategy for the two actions (buy and sell) w.r.t. the inventory z_t for the FQI_2 agent with dashed lines, and the 95% confidence interval, collected with statistical bootstrap, with semi-transparent areas. We observe that the learned policies bid aggressively for large negative and positive inventory. This behavior allows the market maker to keep a zero net inventory on average. This has an important economical interpretation: it shows that the equilibrium policy for the dealers tends to a zero inventory. Note that to reach equilibrium it is necessary to receive negative rewards for adjusting the inventory. Specifically, when ϵ_{sell} and ϵ_{buy} are negative, the agent receives an immediate negative reward with probability 1, due to the fact that it will buy (sell) at price a higher (lower) price than the reference market. FQI_4 provides results in line with the ones just discussed. Note that this behavior, a.k.a. *skewing*, is also a common trait in most market-making strategies designed in the economic literature, e.g., (Guéant 2017).

5.3. Exploitability Study

In this section, we study the setting in which there is a strategic dealer in the market, who has information that allows him/her to exploit the other dealers. This investigation is of paramount relevance to evaluate the performance of the strategies in case

⁶The distribution are truncated so that they have probability 1 of being in the action space $\mathcal{A}$.

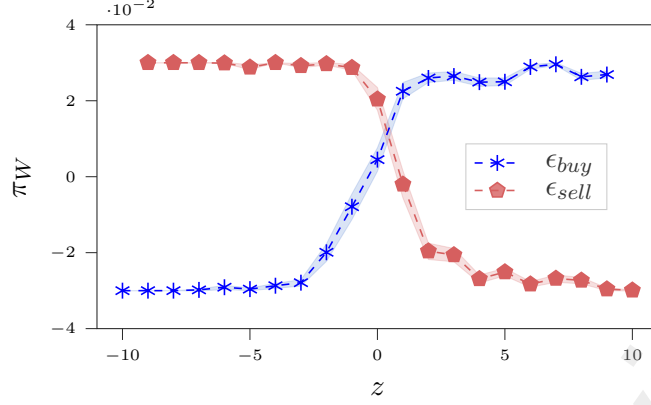


Figure 2.: Equilibrium policy π_W for the FQI_2 agents.

opponents are aware of the currently adopted strategy. Indeed, in a multi-agent setting, there is no concept of *best policy* as its optimality also depends on the aggregate policy of the other players. For instance, when playing a *rock/paper/scissors* game against a player who plays rock more often than scissors, then playing paper is favorable in this situation. On the other hand, this is ruinous in case the other player switches to playing scissors. This is why in a multi-agent framework the correct concept to analyze is *exploitability*, which is commonly used in the game theory literature to evaluate multiple-player settings. Formally, in an N -player stochastic game, the exploitability for the i -th player with policy π^i is defined as:

$$J_{exp}(\pi^i) := \min_{\pi^{-i}} J(\pi^i, \pi^{-i}). \quad (14)$$

Intuitively, the above quantity measures how much value an opponent can gain by using policy π^i . For instance, in the example above (rock/paper/scissors game), assuming at each turn to bet 1\$, the exploitability of playing *rock* with probability 100% is 1\$, i.e., the opponent can make us lose at most 1\$ per round if she/he knows our strategy, while the exploitability of playing rock/paper/scissors each with probability 1/3 is 0\$. To test the exploitability of an agent $\mathfrak{E}$ in our setting, we generate an exploitative agent $E(\mathfrak{E})$ and train it in the following way. We build an environment composed only of agent $\mathfrak{E}$ and agent $E(\mathfrak{E})$. Then, we train agent $E(\mathfrak{E})$ to maximize its own value function. To find an approximately optimal policy we employ an RL algorithm on the MDP which is generated by fixing the behavior of agent $\mathfrak{E}$.⁷ In particular, we used the Proximal Policy Optimization (PPO) policy by Schulman et al. (2017) with a policy network of 2 layers and 64 neurons each. We trained it against the $\mathfrak{E}$ agent described above, creating an exploitative agent $E(\mathfrak{E})$ for each policy described in Section 5.1. We used the implementation of Stable Baselines (Hill et al. 2018) with default parameters for PPO. The provided results have been averaged over 1,000 independent runs.

5.3.1. Exploitability Results

Table 1 reports the performances of the different agents $\mathfrak{E}$ when employed against their corresponding exploitative agent $E(\mathfrak{E})$. We see that the FQI_2 and FQI_4 agents, who are considering the strategic nature of the dealer market setting, perform better

⁷This is a standard approach to approximate the exploitability of the learned policies used by Greenwald et al. (2013).

Table 1.: Performance of the different agents $\mathfrak{E}$ in the setting where the other dealer is the corresponding exploitative agent $E(\mathfrak{E})$. Best results, or tied for the best, are highlighted in boldface.

	FQI_4	FQI_2	Q_4	Q_2	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
L	0.049	0.048	0.0021	-0.02	0.009	0.018	0.026
S	0.008	0.008	0.002	-0.0	0.001	0.002	0.002

Table 2.: Mean dollar reward L for $M_t = 2$. Larger is better.

	FQI_2	Q_2	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_2	0.049	0.059	0.075	0.065	0.061
Q_2	0.03	0.048	0.053	0.044	0.045
$\mathfrak{P}$	0.019	0.052	0.044	0.043	0.048
$\mathfrak{U}$	0.028	0.041	0.042	0.064	0.047
$\mathfrak{N}$	0.04	0.057	0.046	0.048	0.06

than the others when employed against exploitative agents. Indeed, the dollar reward L for FQI_4 and FQI_2 is 0.049 and 0.048, respectively, which is almost twice the second-best agent (Normal $\mathfrak{N}$ with $L = 0.026$). Furthermore, for the same agents, we have a factor 4 improvement in the Sharpe ratio S (0.008 for FQI_4 and FQI_2 vs. 0.002 for $\mathfrak{N}$). The performance improvement is due to the fact that FQI_2 and FQI_4 cannot be manipulated strategically as they have been trained to cope with this kind of adversarial environments. The same is not true for the Q_2 and Q_4 agents, who still consider the strategic aspect of the market-making setting, but are not learning appropriate equilibrium policies. This suggests that the tabular nature of the Q-GMFG algorithm fails to incorporate the inherent regularity of the reward w.r.t. the state and action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$, which, instead, is crucial in training the FQI-GMFG agents.

5.4. Market Simulation Study

In this section, we analyze several market simulations. Specifically, we used the setting of Section 5.1 and for each of the agents described we created a market simulation with one or more other dealer agents. We studied two different scenarios: $M_t = 2$, for all $t \in T$, and $M_t = 4$, for all $t \in T$. For each of the defined metrics, i.e., L , S , and Z , we provided the result obtained by the reference agent (reported on the columns) in a setting where the other market makers are configured as reported on the rows. The provided results have been averaged over 1,000 independent runs. We highlight the best agent over each row in bold.

5.4.1. Market Simulation Results

Table 2 and Table 3 present the average dollar reward L of the agents for $M_t = 2$ and $M_t = 4$, respectively. In particular, when $M_t = 2$ there is always at least one agent

Table 3.: Mean dollar reward L for $M_t = 4$. Larger is better.

	FQI_4	Q_4	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_4, FQI_4, FQI_4	0.03	0.029	0.031	0.031	0.037
Q_4, Q_4, Q_4	0.013	0.016	0.009	0.023	0.033
$\mathfrak{P}, \mathfrak{P}, \mathfrak{P}$	0.019	0.039	0.026	0.035	0.066
$\mathfrak{U}, \mathfrak{U}, \mathfrak{U}$	0.01	0.023	0.015	0.019	0.037
$\mathfrak{N}, \mathfrak{N}, \mathfrak{N}$	0.015	0.011	0.008	0.017	0.021

Table 4.: Mean Sharpe ratio S for $M_t = 2$. Larger is better.

	FQI_2	Q_2	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_2	0.009	0.008	0.01	0.009	0.008
Q_2	0.024	0.007	0.008	0.006	0.007
$\mathfrak{P}$	0.021	0.008	0.006	0.006	0.006
$\mathfrak{U}$	0.023	0.006	0.006	0.009	0.007
$\mathfrak{N}$	0.019	0.008	0.007	0.007	0.009

Table 5.: Mean Sharpe ratio S for $M_t = 4$. Larger is better.

	FQI_4	Q_4	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_4, FQI_4, FQI_4	0.008	0.004	0.004	0.004	0.004
Q_4, Q_4, Q_4	0.014	0.003	0.002	0.003	0.005
$\mathfrak{P}, \mathfrak{P}, \mathfrak{P}$	0.021	0.007	0.004	0.005	0.009
$\mathfrak{U}, \mathfrak{U}, \mathfrak{U}$	0.011	0.002	0.003	0.002	0.006
$\mathfrak{N}, \mathfrak{N}, \mathfrak{N}$	0.011	0.005	0.001	0.003	0.004

that outperforms the FQI_2 one, but there is no clear agent capable of overcoming the others in all settings. Indeed, the Persistent agent $\mathfrak{P}$ provides the largest dollar reward against FQI_2 and Q_2 , the Q_2 agent is the best option against $\mathfrak{P}$ and $\mathfrak{N}$, and the Uniform agent $\mathfrak{U}$ outperforms the others only when tested against an agent of the same kind. Overall, in terms of mean dollar reward, it seems that there is no algorithm able to outperform all the others. Instead, when $M_t = 4$, the Normal agent $\mathfrak{N}$ outperforms all the other agents, with a mean dollar reward L in the range $[0.021, 0.066]$ for all the market configurations considered. However, one cannot base the decision for the best strategy looking at the average dollar reward alone, since large values of this metric might be achieved at the cost of a large risk (in terms of the variance of earnings over time).

Conversely, the results are significantly different if we analyze the Sharpe ratio S , for $M_t = 2$ and $M_t = 4$, as presented in Table 4 and Table 5, respectively. Unlike what has been observed with the mean dollar reward L , the FQI_2 and FQI_4 agents achieve,

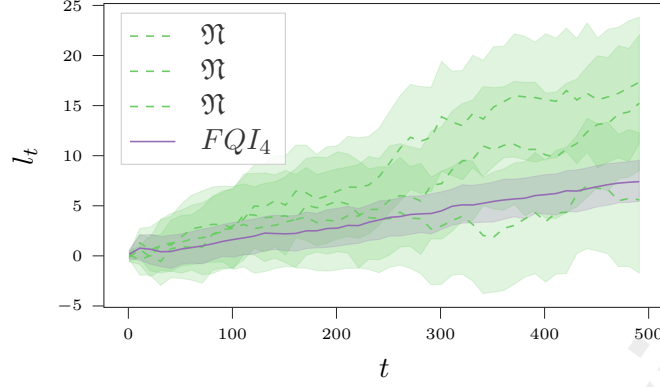


Figure 3.: Average dollar reward l_t of the FQI_4 agent (solid purple line) and those of three identical Normal agents $\mathfrak{N}$ (dashed green lines), when dealing in the same market simulation.

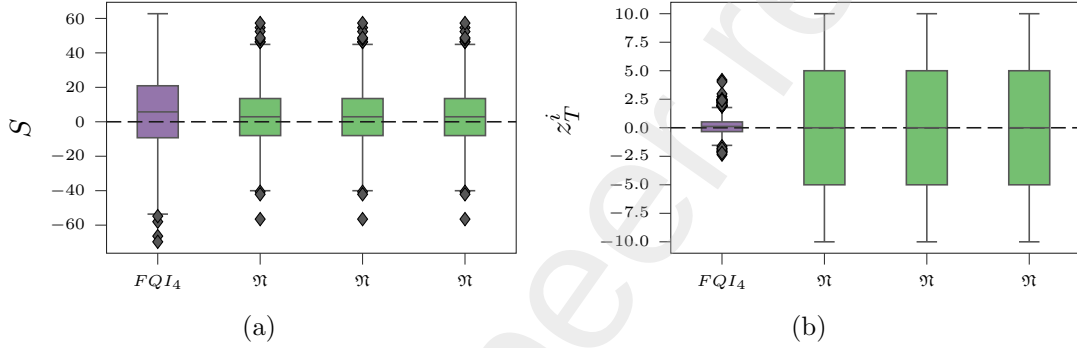


Figure 4.: Box-plot of the distribution over 1,000 runs for (a) Sharpe ratio S ; (b) final inventory z_T^i .

in most scenarios, the best results for the metric S , where FQI_2 is tied in only one setting by the Uniform agent $\mathfrak{U}$. In particular, this happens in the case of $M_t = 2$ as the FQI_2 agent achieves a larger mean Sharpe ratio S than the other agents in all market configurations. Indeed, S is in the range $[0.009, 0.024]$ for FQI_2 while all the other agents have a mean Sharpe ratio S in the range $[0.006, 0.01]$.

On the other hand, when $M_t = 4$, we have that in all the market configurations considered, the FQI_4 agent achieves a higher Sharpe ratio, specifically in the range $[0.008, 0.021]$. By comparing these results with the ones we provided in Table 2 and Table 3, we conclude that the advantage of the Persistent $\mathfrak{P}$, Uniform $\mathfrak{U}$, and Normal $\mathfrak{N}$ agents we saw in terms of mean dollar reward L comes at the cost of high risk, increasing the Sharpe ratio S .

To better understand this phenomenon, we present a detailed example of a specific market configuration for one of the market simulation settings described above.⁸ Figure 3 depicts the dollar reward $l_t, t \in [T]$ in a setting where the FQI_4 agent competes with three Normal agents $\mathfrak{N}$ (i.e., corresponding to the 1-st column and 5-th row of Table 3). The shaded areas are the 95% confidence intervals for the values computed by statistical bootstrap over the 1,000 independent runs of the setting. The results

⁸We opt to provide this level of detail for a single configuration since the behavior we are going to show is common to all of them, therefore the analysis and comments would be the same.

show that the FQI_4 agent achieves positive returns in the market simulation since its trend increases approximately linearly over time. Note that the Normal agents $\mathfrak{N}$ are also generating profit, as the dashed lines corresponding to the average dollar reward are progressively increasing. However, differently from the FQI_4 agent, the Normal agents $\mathfrak{N}$ have a large confidence interval, indicating that they are not reliably achieving profits in each one of the simulations. Conversely, looking at the confidence areas of FQI_4 , we conclude that it consistently provides profits over time. This is due to the specifically crafted reward function used during the training phase, which allows it to maintain a smaller net inventory, thus reducing the standard deviation associated with a large net inventory.

To better investigate this phenomenon, we further analyze the scenario in the 1-st column and 5-th row of Table 3. We report in Figure 4 the box plots associated with the distribution over 1,000 runs of the Sharpe Ratio S (Figure 4a) and the Inventory z_T^i at the final time T (Figure 4b). In Figure 4a the FQI_4 agent has a slightly larger distribution of the Sharpe Ratio S compared to the three Normal agents $\mathfrak{N}$. However, it also has a larger expected value, which is consistent with the results in Table 2 and Table 3. In Figure 4b, the distribution of the final inventory z_T^i is smaller than the ones corresponding to the three Normal agents $\mathfrak{N}$. This is also consistent with the behavior observed in Figure 3. This shows that for all runs, the final inventory z_T^i was small. Recall that having a small inventory is a favorable characteristic of the learned strategy since it prevents market makers from having large capital requirements as stated by the current legislation in most countries.

6. Conclusion and Future Works

In this work, we presented a novel solution to the problem of market-making in dealer markets in the presence of strategic players. The proposed solution is based on the framework of *learning in MFGs*, where the numerical solution is found offline by means of the FQI method. Thanks to this preliminary phase, the strategy learned this way can be executed without requiring further learning processes during the daily operations of the market maker. This approach is capable of handling the competitive nature of the problem, due to the presence of other market makers, by learning an equilibrium strategy in a multi-agent framework.

After presenting the approach, we empirically tested the robustness of the equilibrium strategy in terms of multiple metrics. The experimental evaluation showed that, in the presence of strategic opponents, the method outperforms the other benchmark agents, while in the presence of a generic market configuration, other agents might perform better, as the proposed methods guarantee a *safe* behavior. This shows that the *learning in games* framework can lead to promising outcomes when employed in the market-making context.

An interesting future direction to explore is to add elements of realism to the problem, e.g., by considering a portfolio of correlated assets to trade, or by training on real data. Furthermore, one might consider other RL techniques, such as policy search methods (Williams 1992), which might perform better from an empirical point of view, but at the expense of the theoretical guarantees and/or computational costs. Finally, our approach can be tested/compared with other strategies currently used in a production setting, to check the rationality level, from a game-theoretic point of view, of the players.

References

- Yves Achdou and Italo Capuzzo-Dolcetta. Mean field games: numerical methods. *SIAM Journal on Numerical Analysis*, 48(3):1136–1162, 2010.
- Yves Achdou, Fabio Camilli, and Italo Capuzzo-Dolcetta. Mean field games: numerical methods for the planning problem. *SIAM Journal on Control and Optimization*, 50(1):77–109, 2012.
- Sachin Adlakha, Ramesh Johari, and Gabriel Y Weintraub. Equilibria of dynamic games with many players: Existence, approximation, and market structure. *Journal of Economic Theory*, 156:269–316, 2015.
- Berkay Anahtarci, Can Deha Kariksiz, and Naci Saldi. Fitted q-learning in mean-field games. *arXiv preprint arXiv:1912.13309*, 2019.
- András Antos, Csaba Szepesvári, and Rémi Munos. Fitted q-iteration in continuous action-space mdps. In *Advances in Neural Information Processing Systems*, volume 20, pages 9–16, 2007.
- Robert J Aumann. Markets with a continuum of traders. *Econometrica: Journal of the Econometric Society*, pages 39–50, 1964.
- Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.
- Santiago R Balseiro, Omar Besbes, and Gabriel Y Weintraub. Repeated auctions with budgets in ad exchanges: Approximations and design. *Management Science*, 61(4):864–884, 2015.
- Peter Bank, Ibrahim Ekren, and Johannes Muhle-Karbe. Liquidity in competitive dealer markets. *Mathematical Finance*, 31(3):827–856, 2021.
- Guy Barles and Halil Mete Soner. Option pricing with transaction costs and a nonlinear black-scholes equation. *Finance and Stochastics*, 2(4):369–397, 1998.
- Alain Bensoussan, Jens Frehse, Phillip Yam, et al. *Mean field games and mean field type control theory*, volume 101. Springer, 2013.
- Luis Briceno-Arias, Dante Kalise, Ziad Kobeissi, Mathieu Laurière, A Mateos González, and Francisco J Silva. On the implementation of a primal-dual algorithm for second order time-dependent mean field games with local couplings. *ESAIM: Proceedings and Surveys*, 65: 330–348, 2019.
- Luis M Briceno-Arias, Dante Kalise, and Francisco J Silva. Proximal methods for stationary mean field games with local couplings. *SIAM Journal on Control and Optimization*, 56(2): 801–836, 2018.
- David Byrd, Maria Hybinette, and Tucker Hybinette Balch. Abides: Towards high-fidelity market simulation for ai research. *arXiv preprint arXiv:1904.12066*, 2019.
- Haoyang Cao, Xin Guo, and Mathieu Laurière. Connecting gans and mfgs. *arXiv preprint arXiv:2002.04112*, 2020.
- Pierre Cardaliaguet and Saeed Hadikhanloo. Learning in mean field games: the fictitious play. *ESAIM: Control, Optimisation and Calculus of Variations*, 23(2):569–591, 2017.
- Elisabetta Carlini and Francisco J Silva. A fully discrete semi-lagrangian scheme for a first order mean field game problem. *SIAM Journal on Numerical Analysis*, 52(1):45–67, 2014.
- Elisabetta Carlini and Francisco J Silva. A semi-lagrangian scheme for a degenerate second order mean field game system. *Discrete & Continuous Dynamical Systems*, 35(9):4269, 2015.
- René Carmona and Mathieu Laurière. Convergence analysis of machine learning algorithms for the numerical solution of mean field control and games: II—the finite horizon case. *arXiv preprint arXiv:1908.01613*, 2019.
- René Carmona, Mathieu Laurière, and Zongjun Tan. Linear-quadratic mean-field reinforcement learning: convergence of policy gradient methods. *arXiv preprint arXiv:1910.04295*, 2019a.
- René Carmona, Mathieu Laurière, and Zongjun Tan. Model-free mean-field reinforcement learning: mean-field mdp and mean-field q-learning. *arXiv preprint arXiv:1910.12802*, 2019b.
- Alvaro Cartea, Sebastian Jaimungal, and Jason Ricci. Buy low, sell high: A high frequency

- trading perspective. *SIAM Journal on Financial Mathematics*, 5(1):415–444, 2014.
- Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- Nicholas Tung Chan and Christian Shelton. An electronic market-maker. Technical report, Massachusetts Institute of Technology, 2001.
- Vince Darley, Alexander Outkin, Tony Plate, and Frank Gao. Sixteenths or pennies? observations from a simulation of the nasdaq stock market. In *Proceedings of the IEEE/IAFE/INFORMS Conference on Computational Intelligence for Financial Engineering*, pages 151–154. IEEE, 2000.
- Sanmay Das. A learning market-maker in the glosen–milgrom model. *Quantitative Finance*, 5(2):169–180, 2005.
- Sanmay Das. The effects of market-making on price dynamics. In *Proceedings of the international joint conference on Autonomous agents and multiagent systems*, pages 887–894, 2008.
- Constantinos Daskalakis, Paul W Goldberg, and Christos H Papadimitriou. The complexity of computing a nash equilibrium. *SIAM Journal on Computing*, 39(1):195–259, 2009.
- Damien Ernst, Pierre Geurts, and Louis Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6:503–556, 2005.
- Jean-David Fermanian, Olivier Guéant, and Jiang Pu. The behavior of dealers and clients on the european corporate bond market: the case of multi-dealer-to-client platforms. *Market microstructure and liquidity*, 2(03n04):1750004, 2016.
- Jean-Pierre M Fouque and Zhaoyu Zhang. Deep learning methods for mean field control problems with delay. *Frontiers in Applied Mathematics and Statistics*, 6:11, 2020.
- Zuyue Fu, Zhuoran Yang, Yongxin Chen, and Zhaoran Wang. Actor-critic provably finds nash equilibria of linear-quadratic mean-field games. In *Proceedings of the International Conference on Learning Representations*, 2019.
- Drew Fudenberg, Fudenberg Drew, David K Levine, and David K Levine. *The theory of learning in games*, volume 2. MIT press, 1998.
- Sumitra Ganesh, Nelson Vadori, Mengda Xu, Hua Zheng, Prashant Reddy, and Manuela Veloso. Reinforcement learning for market making in a multi-agent dealer market. *arXiv preprint arXiv:1911.05892*, 2019.
- Diogo A Gomes, Joana Mohr, and Rafael Rigao Souza. Discrete time, finite state space mean field games. *Journal de mathématiques pures et appliquées*, 93(3):308–328, 2010.
- Diogo A Gomes et al. Mean field games models—a brief survey. *Dynamic Games and Applications*, 4(2):110–154, 2014.
- Amy Greenwald, Jiawei Li, Eric Sodomka, and Michael Littman. Solving for best responses in extensive-form games using reinforcement learning methods. In *Proceedings of the conference on Reinforcement Learning and Decision Making*, pages 116–120, 2013.
- Olivier Guéant. *The Financial Mathematics of Market Liquidity: From optimal execution to market making*, volume 33. CRC Press, 2016.
- Olivier Guéant. Optimal market making. *Applied Mathematical Finance*, 24(2):112–154, 2017.
- Olivier Guéant and Iuliia Manziuk. Deep reinforcement learning for market making in corporate bonds: beating the curse of dimensionality. *Applied Mathematical Finance*, 26(5):387–452, 2019.
- Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and financial economics*, 7(4):477–507, 2013.
- Xin Guo, Anran Hu, Renyuan Xu, and Junzi Zhang. Learning mean-field games. In *Advances in Neural Information Processing Systems*, volume 32, pages 4966–4976, 2019.
- Xin Guo, Anran Hu, Renyuan Xu, and Junzi Zhang. A general framework for learning mean-field games. *arXiv preprint arXiv:2003.06069*, 2020.
- Ashley Hill, Antonin Raffin, Maximilian Ernestus, Adam Gleave, Anssi Kanervisto, Rene Traore, Prafulla Dhariwal, Christopher Hesse, Oleg Klimov, Alex Nichol, Matthias Plappert, Alec Radford, John Schulman, Szymon Sidor, and Yuhuai Wu. Stable baselines.

- <https://github.com/hill-a/stable-baselines>, 2018.
- Thomas Ho and Hans R Stoll. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial economics*, 9(1):47–73, 1981.
- Thomas SY Ho and Hans R Stoll. The dynamics of dealer markets under competition. *The Journal of finance*, 38(4):1053–1074, 1983.
- Tin Kam Ho. Random decision forests. In *Proceedings of International Conference on Document Analysis and Recognition*, volume 1, pages 278–282. IEEE, 1995.
- Minyi Huang. Large-population lqg games involving a major player: the nash certainty equivalence principle. *SIAM Journal on Control and Optimization*, 48(5):3318–3353, 2010.
- Minyi Huang, Roland P Malhamé, Peter E Caines, et al. Large population stochastic dynamic games: closed-loop mckean-vlasov systems and the nash certainty equivalence principle. *Communications in Information & Systems*, 6(3):221–252, 2006.
- Minyi Huang, Peter E Caines, and Roland P Malhamé. Large-population cost-coupled lqg problems with nonuniform agents: individual-mass behavior and decentralized *varepsilon*-nash equilibria. *IEEE transactions on automatic control*, 52(9):1560–1571, 2007.
- Krishnamurthy Iyer, Ramesh Johari, and Mukund Sundararajan. Mean field equilibria of dynamic auctions with learning. *Management Science*, 60(12):2949–2970, 2014.
- Petter N Kolm and Gordon Ritter. Dynamic replication and hedging: A reinforcement learning approach. *The Journal of Financial Data Science*, 1(1):159–171, 2019.
- Jean-Michel Lasry and Pierre-Louis Lions. Mean field games. *Japanese journal of mathematics*, 2(1):229–260, 2007.
- Ye-Sheen Lim and Denise Gorse. Reinforcement learning for high-frequency market making. In *Proceedings of the European Symposium on Artificial Neural Networks*, 2018.
- Alex Tong Lin, Samy Wu Fung, Wuchen Li, Levon Nurbekyan, and Stanley J Osher. Apac-net: Alternating the population and agent control via two neural networks to solve high-dimensional stochastic mean field games. *arXiv preprint arXiv:2002.10113*, 2020.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *nature*, 518(7540):529–533, 2015.
- Jun Moon and Tamer Başar. Robust mean field games for coupled markov jump linear systems. *International Journal of Control*, 89(7):1367–1381, 2016.
- Sarah Perrin, Julien Pérolat, Mathieu Laurière, Matthieu Geist, Romuald Elie, and Olivier Pietquin. Fictitious play for mean field games: Continuous time analysis and applications. *arXiv preprint arXiv:2007.03458*, 2020.
- Julia Robinson. An iterative method of solving a game. *Annals of mathematics*, pages 296–301, 1951.
- Lars Ruthotto, Stanley J Osher, Wuchen Li, Levon Nurbekyan, and Samy Wu Fung. A machine learning framework for solving high-dimensional mean field game and mean field control problems. *Proceedings of the National Academy of Sciences*, 117(17):9183–9193, 2020.
- Naci Saldi, Tamer Başar, and Maxim Raginsky. Partially-observed discrete-time risk-sensitive mean-field games. In *Proceedings of the Conference on Decision and Control*, pages 317–322. IEEE, 2019.
- Naci Saldi, Tamer Başar, and Maxim Raginsky. Approximate markov-nash equilibria for discrete-time risk-sensitive mean-field games. *Mathematics of Operations Research*, 45(4):1596–1620, 2020.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Thomas Spooner and Rahul Savani. Robust market making via adversarial reinforcement learning. In *Proceedings of the International Conference on Autonomous Agents and MultiAgent Systems*, pages 2014–2016, 2020.
- Thomas Spooner, John Fearnley, Rahul Savani, and Andreas Koukorinis. Market making via reinforcement learning. In *Proceedings of the International Conference on Autonomous Agents and MultiAgent Systems*, pages 434–442, 2018.

- Hamidou Tembine, Quanyan Zhu, and Tamer Başar. Risk-sensitive mean-field games. *IEEE Transactions on Automatic Control*, 59(4):835–850, 2013.
- Christopher JCH Watkins and Peter Dayan. Q-learning. *Machine learning*, 8(3-4):279–292, 1992.
- Ronald J Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning*, 8(3):229–256, 1992.
- Jiachen Yang, Xiaojing Ye, Rakshit Trivedi, Huan Xu, and Hongyuan Zha. Learning deep mean field games for modeling large population behavior. In *Proceedings of the International Conference on Learning Representations*, 2018a.
- Yaodong Yang, Rui Luo, Minne Li, Ming Zhou, Weinan Zhang, and Jun Wang. Mean field multi-agent reinforcement learning. In *Proceedings of the International Conference on Machine Learning*, pages 5571–5580, 2018b.

Appendix A. Q-GMFG

In this section we describe the algorithm used to train the Q_2 and Q_4 agents used in the experimental section. In particular, in the work by Guo et al. (2020) the authors present an algorithm designed for finite action and state spaces $\mathcal{A}$ e $\mathcal{S}$, that builds on the idea of Algorithm 1. We report it in Algorithm 3 for completeness. The algorithm exploits the Q -learning algorithm to update the approximated equilibrium strategy and uses a simulation approach to evolve the population distribution.

A.1. Algorithm Description

The algorithm works similarly to Algorithm 2, so we highlight here the main differences. Algorithm 3 works by using the Q -learning algorithm to update the Q function. Specifically, it initializes to 0 the Q -function for each state and action (Line 2), then for J steps it either selects the action that currently has the best Q -value with probability $1 - \varepsilon$, or it selects a random action with probability ε (Line 4), a.k.a. ε -greedy algorithm. Using the simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, it extracts the next state s_{t+1} and the reward r_{t+1} , that are used to update of the Q -function (Line 6). After that, the algorithm (namely the update of the population $\mathcal{L}_k$) is identical to Algorithm 2.

A.2. Parameter Setting

For the experimental evaluation of Q_2 and Q_4 , we used $K = 5$ outer iterations and $J = 10 \cdot 10^5$ inner iterations, we fixed the learning rate to $\alpha = 0.8$ and the discount factor to $\gamma = 0.95$.

Algorithm 3 Q-learning for GMFG

Require: Initial state-action distribution $\mathcal{L}_0$, environment simulator $\mathcal{E}(\cdot, \cdot; \mathcal{L})$, number of iterations I, W and J , learning rate α and ε greedy parameter

```

1: for  $k \in [K]$  do
2:   Initialize  $\hat{Q}_{w,0}(s, a) = 0 \ \forall a \in \mathcal{A}, s \in \mathcal{S}$ 
3:   for  $j \in [J]$  do
4:      $a_t \leftarrow \begin{cases} \arg \max_a Q_{w,j}(s_t, a) & \text{with probability } 1 - \varepsilon \\ \text{uniform random} & \text{with probability } \varepsilon \end{cases}$ 
5:      $(s_{t+1}, r_{t+1}) = \mathcal{E}(s_t, a_t; \mathcal{L}_{k-1})$ 
6:      $Q_{w,j+1}(s_t, a_t) = Q_{w,j}(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_{a \in \mathcal{A}} Q_{w,j}(s_{t+1}, a) - Q_{w,j}(s_t, a_t)]$ 
7:   end for
8:   Extract  $\pi_w(s) = \phi_\tau(\hat{Q}_w(s, \cdot)) \ \forall s \in \mathcal{S}$ 
9:    $\mu_w \leftarrow \int_{\mathcal{A}} \mathcal{L}_{w-1}(s, a) da$ 
10:  Initialize  $\mathcal{L}_w(s, a) \leftarrow 0 \ \forall a \in \mathcal{A}, s \in \mathcal{S}$ 
11:  for  $i \in [I]$  do
12:     $s_i \sim \mu_w, a_i \sim \pi_w(s_i)$ 
13:     $s'_i \leftarrow \mathcal{E}(s_i, a_i; \mathcal{L}_w)$ 
14:     $\mathcal{L}_w(s'_i, a_i) \leftarrow \mathcal{L}_w(s'_i, a_i) + 1/I$ 
15:  end for
16: end for

```

Table B1.: Mean reward R for $M_t = 2$. Larger is better.

	FQI_2	Q_2	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_2	-11.203	-19.188	-19.18	-21.129	-20.982
Q_2	-0.287	-16.695	-18.585	-20.271	-19.328
$\mathfrak{P}$	-0.132	-17.526	-17.998	-19.057	-17.809
$\mathfrak{U}$	-0.322	-16.408	-18.697	-17.849	-17.748
$\mathfrak{N}$	-1.245	-17.524	-18.375	-19.153	-18.848

Table B2.: Mean reward R for $M_t = 4$. Larger is better.

	FQI_4	Q_4	$\mathfrak{P}$	$\mathfrak{U}$	$\mathfrak{N}$
FQI_4, FQI_4, FQI_4	-8.617	-21.076	-20.473	-19.996	-21.521
Q_4, Q_4, Q_4	-0.127	-18.001	-19.951	-18.028	-17.909
$\mathfrak{P}, \mathfrak{P}, \mathfrak{P}$	-0.162	-17.355	-19.137	-17.999	-17.998
$\mathfrak{U}, \mathfrak{U}, \mathfrak{U}$	-0.141	-18.438	-17.66	-17.611	-18.085
$\mathfrak{N}, \mathfrak{N}, \mathfrak{N}$	-0.333	-17.562	-18.776	-17.62	-18.638

Appendix B. Market Simulation Study: Reward

In this section we provide additional result on the market simulation part of the experimental investigation, which we presented in Section 5.4. Table B1 and Table B2 presents the average reward:

$$R := \sum_{t \in [T]} \frac{r_t}{T},$$

where the reward r_t is defined in Equation (4). We can see that in almost all the instances, the FQI_2 and FQI_4 agents perform better than the other agents. For instance, in the case $M_t = 2$, over all the configurations, the FQI_2 has a mean reward R of ≈ -11 and ≈ -0.1 against the FQI_2 and Q_2 agents, while all the other agents have a mean reward of ≈ -19 . Similarly, in the case of $M_t = 4$ we observe that the FQI_4 agent has a mean reward of in the range $[-8.6, -0.3]$, in all the configurations, while the other agents have a mean reward of ≈ -20 in all configurations. We remark that this table is presented only for completeness, as the agents $\mathfrak{P}$, $\mathfrak{U}$, $\mathfrak{N}$ are not explicitly optimizing the metric R , while the strategic agents are optimizing this value in their learning procedure. However, such table shows that the Algorithm 3 fails to learn optimal policies due to the large sizes of the Q -tables involved. This justifies the introduction of Algorithm 2, which solves this problem by using function approximation tool on the Q -tables, namely regression trees.